

SHAP Analysis Report

Employee Attrition Prediction Model

1. Objective

The objective of this analysis is to identify the key factors influencing employee attrition and explain the machine learning model's predictions using SHAP (SHapley Additive exPlanations).

2. SHAP Interaction Analysis

The SHAP interaction plot was generated to understand how different features interact with each other in influencing attrition predictions.

Features Analyzed:

- Age
- BusinessTravel

Observations:

- Most SHAP interaction values are concentrated around zero, indicating a relatively weak interaction effect between Age and BusinessTravel.
- Employee age alone does not appear to strongly influence attrition risk when combined with business travel frequency.
- Business travel contributes independently to attrition prediction, but its interaction with age is limited.
- A small number of observations show moderate positive and negative interaction values, suggesting that for specific employees the combination of age and travel requirements may slightly increase or decrease attrition risk.

3. Key Findings

- Age is not a primary driver of employee attrition in the current model.
- Business travel has some influence on attrition but interacts minimally with employee age.
- The majority of attrition predictions are likely driven by other factors such as:
 - Overtime
 - Monthly Income
 - Job Satisfaction
 - Work-Life Balance

- Years Since Last Promotion

4. Business Interpretation

The analysis suggests that HR teams should focus more on workload management, compensation, career growth opportunities, and employee satisfaction rather than age-based interventions.

Employees who frequently travel for work may require additional support; however, age alone should not be considered a significant risk indicator for attrition.

5. Conclusion

SHAP interaction analysis indicates that the Age–BusinessTravel relationship has a limited impact on attrition predictions. Organizational efforts to reduce employee turnover should prioritize factors with stronger predictive influence identified through feature importance and SHAP summary analysis.